PHY201: Waves & Optics

Solutions : 1st Mid Sem

August 31, 2025

1. $m = 1.00$ kg, $k = 100\ \mathrm{N\,m^{-1}}$, $x(0) = x_0 = 0.030$ m, $\dot{x}(0) = v_0 = -0.40\ \mathrm{m\,s^{-1}}$.

   (a) *Equation and ω_0.*

   $$m\ddot{x} + kx = 0 \iff \ddot{x} + \omega_0^2 x = 0, \qquad \omega_0 = \sqrt{\frac{k}{m}} = \sqrt{\frac{100}{1}} = 10\ \mathrm{s^{-1}}.$$

   (b) *Amplitude–phase form.* Write $x(t) = A\cos(\omega_0 t + \phi)$ with

   $$A = \sqrt{x_0^2 + \left(\frac{v_0}{\omega_0}\right)^2} = \sqrt{(0.03)^2 + \left(\tfrac{-0.40}{10}\right)^2} = \sqrt{0.0009 + 0.0016} = 0.050\ \mathrm{m}.$$

   Using $x_0 = A\cos\phi$ and $v_0 = -A\omega_0 \sin\phi$,

   $$\cos\phi = \frac{0.03}{0.05} = \frac{3}{5}, \qquad \sin\phi = \frac{0.40}{0.05\cdot 10} = \frac{4}{5} \Rightarrow \phi = \arctan\!\left(\frac{4}{3}\right).$$

   Hence

   $$x(t) = 0.050\cos\!\left(10\,t + \arctan\frac{4}{3}\right)\ \mathrm{m}\,.$$

   (c) *Total energy.*

   $$E = \frac{1}{2}kA^2 = \frac{1}{2}\cdot 100\cdot (0.05)^2 = \frac{1}{2}\cdot 100 \cdot 0.0025 = 0.125\ \mathrm{J} = \frac{1}{8}\ \mathrm{J}\,.$$

2. $\ddot{x} + \gamma\dot{x} + \omega_0^2 x = 0$ with $\omega_0 = 20\pi\ \mathrm{s^{-1}}$.

   (a) Amplitude envelope is $A(t) \propto e^{-(\gamma/2)t}$.
   Amplitude drops by a factor e in $N = 40$ cycles $\Rightarrow$ over $t \approx NT_0 = N\cdot\frac{2\pi}{\omega_0}$:

   $$e^{-(\gamma/2)t} = e^{-1} \;\Rightarrow\; \frac{\gamma}{2}t = 1 \;\Rightarrow\; \gamma = \frac{2}{t}.$$

   With $t = 40\cdot\frac{2\pi}{20\pi} = 4$ s:

   $$\gamma = \frac{2}{4} = 0.5\ \mathrm{s^{-1}}\,.$$

   (b) $Q = \dfrac{\omega_0}{\gamma}$:

   $$Q = \frac{20\pi}{0.5} = 40\pi\,.$$

   (c)

   $$\omega = \sqrt{\omega_0^2 - \left(\frac{\gamma}{2}\right)^2} = \sqrt{(20\pi)^2 - \left(\tfrac{1}{4}\right)^2} = \sqrt{400\pi^2 - \tfrac{1}{16}}\ \mathrm{s^{-1}}\,.$$

   Comment: Since $(\gamma/2)^2 = \frac{1}{16} \ll 400\pi^2$, the shift $\omega_0 - \omega$ is negligible.

3. Given $Q = 50$ and $f = 5$ Hz, so $\omega = 2\pi f = 10\pi$ s^{-1}. Energy decays as $E(t) = E_0 e^{-\gamma t}$ where $\gamma = \omega/Q = \dfrac{10\pi}{50} = \dfrac{\pi}{5}$ s^{-1}. One cycle has $T = \dfrac{2\pi}{\omega} = \dfrac{1}{5}$ s.

   (a) *Fractional energy loss per cycle.*

$$\frac{\Delta E}{E} = 1 - e^{-\gamma T} = 1 - e^{-\pi/25} \;\approx\; \gamma T = \frac{\pi}{25} \quad \text{(small damping)}.$$

   (b) *Amplitude ratio after $N = 25$ cycles.* Amplitude envelope $A(t) \propto e^{-(\gamma/2)t}$ gives

$$\frac{A_N}{A_0} = e^{-(\gamma/2)NT} = e^{-(\pi/10)\cdot 25\cdot(1/5)} = e^{-\pi/2}\,.$$

4. Data: $m = 0.25$ kg, $b = 2$ N s m^{-1}, $k = 25$ N m^{-1}, $F_0 = 10$ N, $\omega = 10$ s^{-1}. Steady state $x(t) = A\cos(\omega t - \delta)$.

   (a) *Amplitude and phase.*

$$A = \frac{F_0}{\sqrt{(k - m\omega^2)^2 + (b\omega)^2}}, \qquad \tan\delta = \frac{b\omega}{k - m\omega^2}.$$

   Here $k - m\omega^2 = 25 - 0.25 \cdot 100 = 0$, hence

$$A = \frac{F_0}{b\omega} = \frac{10}{2 \cdot 10} = \frac{1}{2}\ \text{m}, \qquad \delta = \frac{\pi}{2}\,.$$

   (b) *Energy dissipated per cycle.*

$$E_{\text{cycle}} = \int_0^T b\,\dot{x}^2\,dt = \pi\, b\, A^2\,\omega = \pi \cdot 2 \cdot \left(\frac{1}{2}\right)^2 \cdot 10 = 5\pi\ \text{J}\,.$$

   (c) *Mean power input.*

$$\overline{P} = \frac{E_{\text{cycle}}}{T} = \frac{5\pi}{2\pi/\omega} = \frac{5 \cdot 10}{2} = 25\ \text{W} \quad \big(\text{equivalently } \overline{P} = \tfrac{1}{2}\, b\, A^2\, \omega^2\big).$$